

EDUCATION	Princeton University Ph.D. in Physics, Research Advisor:	<i>as an NSF Graduate Research Fellow</i> <i>Herman L. Verlinde</i>	<i>August 2023 - Present</i>
	University of Pennsylvania Master of Science in Physics, Bachelor of Arts in Physics, Minor in Mathematics, Research Advisor:	<i>concentrating in Astrophysics</i> <i>James M. Kikkawa</i>	<i>August 2019 - May 2023</i>
RESEARCH INTERESTS	I am interested in theories of quantum gravity that explore black hole evaporation and disappearance. In particular, I study toy models of quantum gravity in 1+1 dimensions to analyze if and how unitarity can coexist with processes that change the spacetime topology. My current work focuses on the emergence of a semiclassical Hilbert space associated to closed universes nucleated within the double-scaled Maldacena-Qi deformation of the SYK model.		
RESEARCH SUMMARY	Graduate Student with Prof. Herman Verlinde Princeton University Department of Physics		<i>August 2023 - Present</i>
	- Derived chord rules for computing the partition function of a holographic spin chain model (DSSYK-MQ) in multiple distinct bulk topological phases. Used these to compute microscopic semiclassical Hartle-Hawking states. - Proved the existence of a higher topology saddle point in that same model dual to a bulk spacetime containing a nontrivial baby universe.		
	Student Researcher with Prof. James Kikkawa UPenn Department of Physics and Astronomy		<i>May 2021 - August 2023</i>
	- Developed a novel technique for generating beams of light with stable phase singularities, thus carrying consistent and focused quantized orbital angular momentum. - Modeled the optical field diffraction of a grating with arbitrary transmission and retardance, and refraction through a lens with arbitrary focal length, using C++ code I developed.		
	College Intern with Dr. George Clark, Dr. Peter Kollmann Johns Hopkins University Applied Physics Lab		<i>June 2018 - January 2022</i>
	- Modeled plasma loss processes in Saturnian magnetosphere using Python code I constructed, with initial conditions drawn from measured Cassini CHEMS data. - Showed using this model that charge exchange is the dominant loss process for hydrogen ions and water group ions near Saturn in the keV range.		
PUBLICATIONS & PREPRINTS	[3] AS and Herman Verlinde, <i>Baby Universe in a Coupled SYK Model</i> , 2605.05291 .		
	[2] AS , Mehmet A. Noyan, and James M. Kikkawa, <i>High purity orbital angular momentum of light</i> , Optics Express 30 , 24 (2022).		
	[1] AS , George Clark, and Peter Kollmann, <i>Charge exchange ion losses in Saturn's magnetosphere</i> , Journal of Geophysical Research: Space Physics 126 , 10 (2021).		

PRESENTATIONS & CONFERENCES	[4] <i>Baby Universe in a Coupled SYK Model</i> Julius-Maximilians-Universität Würzburg, Würzburg, Germany, 2 June 2026.
	[3] <i>Charge Exchange Ion Losses in Saturn's Magnetosphere</i> American Geophysical Union Fall Meeting, New Orleans, LA, 15 December 2021.
	[2] <i>Orbital Angular Momentum: Huygens-Fresnel GPU Simulation</i> University of Pennsylvania Interdisciplinary Group Meeting, Philadelphia, PA, 22 October 2021.
	[1] <i>Charge Exchange Ion Losses in Saturn's Magnetosphere</i> Outer Planets Assessment Group, (Remote), 31 August 2021.
TEACHING EXPERIENCE	Teaching Assistant, Physics Princeton University Department of Physics <i>August 2025 - Present</i> • Spring 2026: Physics 104 – General Physics II (Electromagnetism) • Fall 2025: Physics 305 – Quantum Mechanics
	Teaching Assistant, Mathematics University of Pennsylvania Department of Mathematics <i>August 2021 - August 2023</i> • Spring 2023: Math 114 – Multivariable Calculus • Fall 2022: Math 104 – Introductory Calculus • Spring 2022: Math 104 – Introductory Calculus • Fall 2021: Math 104 – Introductory Calculus
	Tutor, Physics and Mathematics <i>August 2020 - Present</i>
	Princeton Prison Teaching Initiative Tutor, Mathematics NJ-STEP, Garden State Youth Correctional Facility <i>January 2026 - Present</i> • Co-lead 1-on-1 and group sessions focused on remedial and collegiate mathematics for incarcerated students.
ACADEMIC OUTREACH	Graduate Student Mentor, Physics Princeton University Department of Physics <i>August 2024 - Present</i> • Guided 7 students majoring in physics by discussing how to reach goals through course selection, involvement in organizations, participation in research, and applications for scholarships and jobs. • Reviewed and gave feedback on scholarship, research, and summer internship applications.
	National Science Foundation Graduate Research Fellowship <i>April 2023</i> • One of 2254 graduate students (175 in physics and astronomy) in the United States selected from a pool of over 13000 students to receive an NSF GRFP Fellowship, on the basis of: “excellent intellectual merit and broader impacts.” • “The NSF GRFP recognizes and supports outstanding graduate students in STEM disciplines who are pursuing research-based doctoral degrees at accredited US institutions.” (citation)
SELECT AWARDS	Roy and Diana Vagelos Science Challenge Award <i>July 2022</i> • One of 7 students in the Penn STEM program, only 2 of whom are physics majors, to receive a scholarship fully covering all tuition and expenses at Penn until graduation, independent of financial need.
COMPUTER SKILLS	I have extensive programming experience in Python , including computations and data analysis with NumPy, SciPy, and AstroPy, as well as data visualization with Matplotlib. Additionally, I can write and run optimized code for graphic processors in C++ using the CUDA package. I am also fluent in Mathematica , MATLAB , L^AT_EX , and TikZ. I have a working knowledge of HTML and CSS .